

Problem Statement

LLM assistants forget user preferences stated earlier in a conversation (PrefEval, ICLR 2025): accuracy collapses as unrelated context grows. Project: a trained preference-detection classifier + preference memory + modern retrieval. Mid-term covers Phase 1.

1) **Group Name:** Individual project

2) **Team Name, Roll no:** Krishna Jhanwar, 12341250

3) **Novelty:**

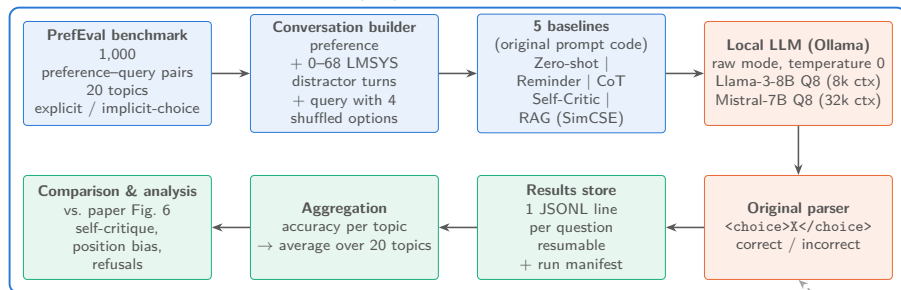
- **Overall idea:** a small *trained* classifier detects preference-stating turns and writes them to a structured memory (grows with the number of preferences, not turns), combined with a modern retriever (bge / e5, replacing 2021 SimCSE)
- **Phase 1 (done):** PrefEval's Bedrock-only evaluation rebuilt to run fully locally (Ollama, one 12 GB GPU) with the original prompts unchanged: resumable, logged, reproducible
- **Phase 1:** baselines reproduced on two of the paper's models (Llama-3-8B, Mistral-7B): 5 methods, explicit + implicit-choice, 0.2k–23k tokens, 64,535 questions; new implicit classification baselines
- **Phase 1:** found and fixed 3 prompt-formatting issues and 2 defects in the released retrieval data; new analyses (self-critique overturning answers, position bias at long context, RAG refusals)

4) **Contributions:**

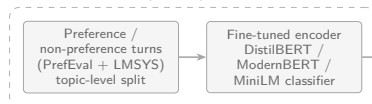
Member 1 — Krishna Jhanwar (12341250): project design (all phases); Phase 1: local evaluation pipeline, debugging of prompt and data issues, all baseline runs (≈ 71 GPU-hours), aggregation, analysis and reporting

Architecture

Phase 1 — evaluation harness & baseline reproduction (done)



Phase 2 — classifier (planned)



Phase 3 — classifier + memory + retrieval (planned)



Results and Findings in Tabular Format

Explicit preference: classification accuracy (%), averaged over 20 topics. Paper (left) vs. our reproduction (right)

Model	Method	Paper (Fig. 6, Bedrock)						Ours (local, Ollama Q8)					
		0.2k	1k	3k	10k	16k	23k	0.2k	1k	3k	10k	16k	23k
Llama-3-8B	Zero-shot	85	55	42	–	–	–	94.2	52.7	43.3	–	–	–
	Reminder	94	80	59	–	–	–	97.4	78.0	59.4	–	–	–
	CoT	83	67	50	–	–	–	93.7	78.0	63.9	–	–	–
	Self-Critic	79	73	62	–	–	–	78.8	50.1	46.7	–	–	–
	RAG	–	87	80	–	–	–	–	88.6	74.8	–	–	–
Mistral-7B	Zero-shot	87	66	57	50	42	47	95.5	61.5	55.5	42.5	37.1	37.6
	Reminder	95	85	75	63	56	56	96.2	78.9	71.3	56.7	43.9	46.9
	CoT	90	79	73	69	56	62	88.0	74.9	67.0	–	–	–
	Self-Critic	75	65	58	51	52	48	79.8	67.0	61.0	–	–	–
	RAG	–	87	86	81	80	75	–	81.0	75.7	64.6	59.0	53.7

– = not available: Llama-3-8B has an 8k context limit (as in the paper); RAG is undefined at 0.2k; our CoT and Self-Critic were run up to 3k only.

Explicit vs. implicit (choice) at 3k tokens (%)

Method	Llama-3-8B		Mistral-7B	
	Explicit	Implicit	Explicit	Implicit
Zero-shot	43.3	46.0	55.5	47.5
Reminder	59.4	60.2	71.3	63.1
CoT	63.9	61.7	67.0	59.5
Self-Critic	46.7	38.7	61.0	50.9
RAG	74.8	73.7	75.7	64.0

Key findings

Finding	Evidence
Baselines reproduced locally	12/29 cells within 5 pts; 20/29 within 10
Accuracy decays with context	Mistral Zero-shot 95.5 → 37.6 (0.2k → 23k)
RAG most robust at long context	23k: RAG 53.7 vs. Zero-shot 37.6
Self-critique overturns answers	Llama, 0.2k: 205 right→wrong vs. 51 wrong→right
Long context → position prior	23k: accuracy 42 (answer A) vs. 23 (answer D)
Implicit preferences harder	Mistral: 7–15 pts below explicit
Retrieval triggers refusals	Llama RAG, 3k: 19.1% without a valid choice